

RepSync: Real-Time Form Correction and Feedback in Virtual Benchpress Training

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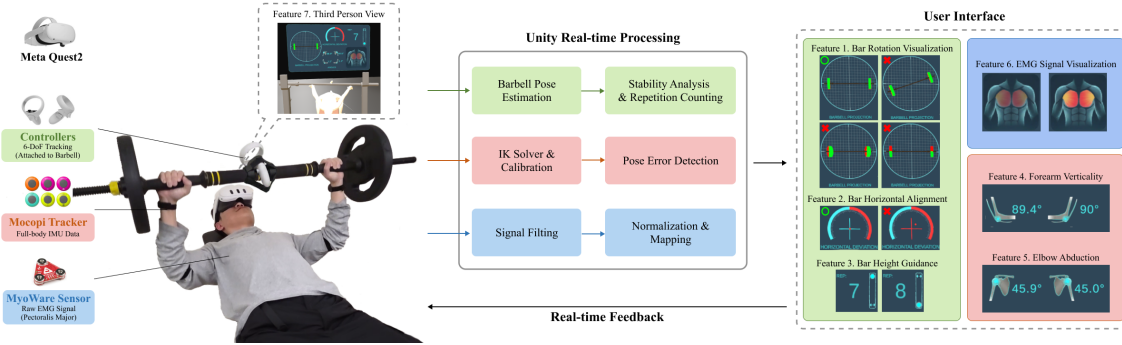


Figure 1: System pipeline

ABSTRACT

Improper bench press posture increases injury risk and reduces training efficiency. We present RepSync, a VR-based coaching system providing real-time multimodal feedback for posture correction. By integrating wearable Inertial Measurement Units (IMUs) and wireless EMG, RepSync monitors limb kinematics and muscle activation. The system evaluates form based on forearm verticality and elbow abduction, delivering corrective visual cues via an immersive dashboard. By projecting physiological data into the user's field of view, RepSync enables self-correction of biomechanical errors without external supervision. Preliminary results indicate that the system effectively mitigates common technical faults for both novice and experienced lifters.

Index Terms: VR training, Embodied Interaction, Posture Correction, Multimedia Feedback.

1 INTRODUCTION

The bench press is a fundamental strength training exercise that requires precise upper-body coordination. Incorrect form such as excessive elbow abduction, improper barbell position, or asymmetric extension significantly increases the risk of injury. While traditional coaching provides essential oversight, it is often unavailable for home-based training. Existing digital solutions often rely on computer vision [2], which suffers from self-occlusion during supine exercises. To address these limitations, we present RepSync,

a VR-based training system designed for general resistance-training users that prioritizes real-time postural correctness and physiological awareness. Our system is designed around three core contributions: (1) Occlusion-Resilient Tracking: Utilizing a streamlined wearable configuration to maintain high-fidelity posture data where camera-based systems sometimes fail due to occlusion. (2) Multimodal Feedback Integration: Synchronizing kinematic trajectory cues with EMG-driven muscle activation heatmaps. (3) Real-time Adjustment: Providing an "in-situ" immersive dashboard that allows trainees to visualize and correct their form in real-time. By fusing these core functionalities, RepSync enables users to develop a more accurate mental model and improved motor learning of correct bench press mechanics.

2 IMPLEMENTATION

Our VR bench press assistance system (Fig. 1) integrates motion capture, barbell tracking, and muscle activity sensing to provide real-time training feedback. The system is built in Unity and uses a Meta Quest 2 headset with controllers, Sony Mocopi sensors, and a MyoWare 2.0 Muscle Sensor Wireless Kit. Our feedback design follows biomechanics findings that EMG alone is insufficient, and that grip width/elbow positioning and lateral bar forces affect joint loading and stability [3]; therefore we provide barbell stability cues (Features 1–3), monitor elbow kinematics (Features 4–5), and visualize muscle activation.

2.1 Sensing and Data Acquisition

RepSync utilizes a multimodal sensor set integrated via Unity 2022.3.62f1. Barbell tracking is achieved using a **Meta Quest 2** controller rigidly mounted to the bar's center to provide 6-DoF pose data. For body kinematics, **Sony Mocopi** IMUs are placed on the head, pelvis, upper arms, and wrists, with full-body poses streamed to Unity via UDP after an initial IK-based calibration. Physiological feedback is captured via a **MyoWare 2.0** wireless EMG sensor on the chest, transmitting raw muscle activation signals over Bluetooth for real-time visualization.

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2.2 Calibration and Reference Pose Capture

Before training begins, the system performs a user-specific calibration to establish a biomechanical baseline. Rather than using a one-size-fits-all model, we record a **Reference Starting Pose** unique to the user's limb lengths and the physical bench dimensions. Once the starting pose is validated, the system records the 3D coordinates of the barbell. All subsequent real-time visualization feedback is calculated relative to this captured baseline.

2.3 Visualization and Feedback Module

After readiness is confirmed, the user is teleported to a dedicated visualization environment with the following visualization features.

Barbell metrics (Features 1–3). Bench press biomechanics highlights that technique variations alter joint loading and that lateral components of barbell interaction are meaningful for understanding stability and mechanical demand [3]. Accordingly, we provide real-time barbell-centric cues derived from the 6-DoF bar pose, including (1) *barbell rotations* to indicate stability and unintended twisting/tilting, (2) *barbell horizontal alignment* (center drift/offset) to promote a consistent bar path, and (3) *bar height guidance* to support repeatable range of motion and tempo (see Fig. 1, Features 1–3).

Postural monitoring (Features 4–5). Because grip width and elbow positioning modulate shoulder vs. elbow loading, monitoring elbow kinematics provides actionable technique feedback [3]. Using Mocopi skeletal data, we compute and visualize (i) elbow abduction angle and (ii) elbow flexion angle to detect common technical faults (see Fig. 1, Features 4–5). These cues complement barbell metrics: the bar pose captures outcome-level stability, while elbow angles reveal posture-level causes.

EMG & Perspective (Features 6–7): Chest engagement is visualized via an intensity map on a synthetic model. Additionally, an optional third-person view renders the user and barbell simultaneously to clarify spatial relationships often obscured in first-person perspectives.

Overall, the system provides synchronized visual feedback of barbell kinematics, postural correctness, and EMG-based muscle activation to assist users in maintaining stability, consistent bar path, and effective muscle engagement during bench press training.

3 EVALUATION

We conducted experiments to evaluate RepSync's usability, comfort, and training effectiveness with 16 participants (13 male, 3 female; age 19–28; height 159–184 cm; weight 49–85 kg). 10 participants reported prior VR experience, while 6 reported none. Benchpress experience varied: 4 reported no experience, 5 reported less than one year, and 7 reported more than one year. Regarding prior discomfort experience during VR headset use or 3D gameplay, 4 reported never, 8 reported occasional, and 4 reported frequent.

Training Effectiveness. Participants rated RepSync's seven core features and overall usefulness on an integer 0–10 scale. The results are summarized in Tab. 1. Participants were grouped by self-reported benchpress experience: *Novice* (less than 1 year) and *Experienced* (more than 1 year). Overall, RepSync achieved a mean score of 8.75, suggesting that the system delivers meaningful and intuitive training feedback. Both groups rated *Bar Rotation Visualization* and *Third Person View* as the most preferred features. Across all features, novice participants provided consistently higher ratings than experienced participants, with the largest group difference observed for *EMG Signal Visualization*.

Usability. RepSync achieved a mean System Usability Score (SUS) [1] score of 68.9, indicating acceptable usability. Despite the burden of wearing multiple sensors and the system's complex setup, participants rated the item "I would like to use this system frequently" at 4.31, suggesting that the form-correction feature outweighed these inconveniences. Post-study interviews highlighted

Table 1: Mean Feature Ratings: Novice vs. Experienced Users

Feature	Novice	Experienced	All
Bar Rotation Visualization	9.00	9.00	9.00
Bar Horizontal Alignment	8.22	6.71	7.56
Bar Height Guidance	8.33	8.28	8.31
Forearm Verticality	8.77	7.14	8.06
Elbow Abduction	8.77	7.42	8.18
EMG Signal Visualization	8.44	6.71	7.68
Third Person View	9.00	8.71	8.87
Overall Rating	9.00	8.42	8.75

Table 2: Analysis of SSQ

Measure	Never	Occasional	Frequent	All
Nausea	7.15	11.92	11.92	10.73
Oculomotor	17.05	20.84	24.63	20.84
Disorientation	13.92	26.10	34.80	25.23
Overall Score	14.96	21.97	26.18	21.27

needs for a smoother user flow to reduce technician reliance, clearer UI explanations, and the integration of a passthrough mode to enhance user safety.

Comfort. The average Simulator Sickness Questionnaire (SSQ) score was 21.27, indicating moderate simulator sickness. Detailed results are reported in Tab. 2. Participants were grouped by prior experience of discomfort during VR headset use or 3D gameplay (*Never, Occasional, Frequent*). Disorientation emerged as the dominant contributor to SSQ scores, potentially driven by viewpoint changes in VR. To mitigate this effect, the *Third Person View* feature could be made toggleable, allowing users to switch to a comfort-preserving mode when needed. Across all three groups, nausea ratings remained relatively low. In post-study interviews, one participant reported that simultaneous presentation of multiple feedback cues induced eye fatigue.

4 CONCLUSION

We presented **RepSync**, a VR system that facilitates bench press form correction through real-time feedback. By integrating barbell-centric pose tracking with EMG-driven muscle visualization, RepSync reframes strength training as a trajectory-guided interaction rather than a complex motion-capture problem. Our future work will focus on data-driven trajectory optimization and longitudinal studies to evaluate the system's impact on long-term injury prevention.

ACKNOWLEDGMENTS

This research was supported by the National Science and Technology Council (NSTC 114-2223-E-007 -012 -MY3 and NSTC 112-2221-E-007-079-MY3), Taiwan.

REFERENCES

- [1] J. Brooke. Sus: A 'quick and dirty' usability scale. In P. W. Jordan, B. Thomas, I. L. McClelland, and B. Weerdmeester, eds., *Usability Evaluation in Industry*, pp. 189–194. Taylor & Francis, London, 1996.
- [2] Y.-M. Ko, A. Nasridinov, and S.-H. Park. Real-time ai posture correction for powerlifting exercises using yolov5 and mediapipe. *IEEE Access*, 2024. 1
- [3] L. Mausehund, A. Werkhausen, J. Bartsch, and T. Krosshaug. Understanding bench press biomechanics—the necessity of measuring lateral barbell forces. *The Journal of Strength & Conditioning Research*, 36(10):2685–2695, 2022. 1, 2